Database Management System (DBMS)

Previous Year Question Paper – 2024
Time: 3 Hours Maximum Marks: 100

Note: This is a sample/dummy placeholder question paper generated for demonstration purposes only. It does not represent an official examination paper. Some questions repeat across years with different wording, as is common in real previous-year question compilations.

Section A – Short Answer Questions (Answer any 5)

Q1. What is the purpose of a trigger in SQL? Write a simple example. *(5 marks each)*

Q2. Explain the concept of serializability with respect to concurrent transactions. *(5 marks each)*

Q3. What is denormalization? Mention its advantages and disadvantages. *(5 marks each)*

Q4. Write the general syntax of SELECT with GROUP BY and HAVING, and explain with an example. *(5 marks each)*

Q5. What is the purpose of an ER diagram in database design? *(5 marks each)*

Section B – Medium Answer Questions (Answer any 4)

Q1. Discuss the characteristics of a data warehouse and its typical architecture. *(10 marks each)*

Q2. What is the Three-Schema Architecture? Explain each level with a diagram. *(10 marks each)*

Q3. Describe the role of different keys in maintaining integrity of a relational database. *(10 marks each)*

Q4. Design an Entity-Relationship model for a Library Management System with at least four entities. *(10 marks each)*

Section C – Long Answer Questions (Answer any 3)

Q1. Explain how a DBMS optimizes a query, covering query trees and cost estimation. *(15 marks each)*

Q2. Model a University Management System with Students, Courses and Faculty, and write sample SQL queries for common operations. *(15 marks each)*

Q3. What is data abstraction in DBMS? Explain its levels with examples. *(15 marks each)*

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